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SPOTIFY TRACK POPULARITY ANALYTICS

*A Data Preparation, Cleaning and Statistical Analysis Pipeline for Uncovering
the Drivers of Track Popularity on Spotify*

SEMESTER-I

DATA PREPARATION AND DATA MANAGEMENT

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1.INTRODUCTION

Music streaming platforms such as Spotify generate rich, multi-dimensional metadata for every track they host — audio metadata, artist metrics, album context, release history, and engagement-derived popularity scores. Understanding what actually drives a track's popularity score is valuable to artists, labels, playlist curators, and the platform itself, yet the raw data behind such analysis is rarely analysis-ready: it contains missing fields, duplicate or repeated releases, inconsistent data types, and skewed distributions that can mislead naive analysis.

This project builds an end-to-end data preparation and analytics pipeline on a dataset of 8,582 raw Spotify tracks (reduced to 8,579 after cleaning). The pipeline covers data quality assessment, cleaning and preprocessing, feature engineering (including title sentiment analysis), descriptive and inferential statistics, and an interactive live dashboard, in order to answer a single central question: what measurable factors are associated with a track being more or less popular on Spotify?

1.1 Objectives

- To assess and improve the quality of a raw Spotify tracks dataset using a combination of AI-assisted diagnostic review and manual validation.
- To design and implement a reproducible cleaning and preprocessing pipeline covering missing values, duplicates, outliers, and data type integrity.
- To engineer new analytical features, including sentiment scores derived from track titles and flags for repeat releases and missing follower data.
- To identify statistically significant drivers of track popularity using correlation analysis, feature selection, and hypothesis testing (t-test, ANOVA, chi-square).
- To visualize distributional, categorical, relational, and time-based patterns in the cleaned dataset.
- To publish the findings through a live, filterable analytics dashboard for interactive exploration.

1.2 Problem Statement

Raw platform-exported datasets — including this Spotify track catalogue — routinely suffer from incomplete metadata (e.g., unmapped genres), latent duplication (the same song re-released across albums, singles, and compilations), skewed numerical distributions (a small number of mega-popular artists dominate the follower count distribution), and mixed or loosely-typed columns. These issues make it unsafe to draw conclusions directly from the raw export.

The central problem this project addresses is: **how can a raw, imperfect Spotify tracks dataset be systematically cleaned and validated, and then analyzed, to reliably identify the factors that are statistically associated with track popularity?**

Answering this helps artists and labels understand which levers (explicit content, album format, artist reach, release timing, track length) are worth optimizing for, and gives analysts a validated, reusable data preparation workflow for future streaming datasets.

1.3 Project Assets

The full analytical workflow and an interactive dashboard built on top of the analysis are available as follows. The dashboard mirrors the statistics computed in the notebook (8,579 tracks, 13 analytical features) and allows live filtering by genre, album type, explicit flag, release year, and popularity range.

Analysis Notebook: Spotify_DPDM.ipynb (submitted alongside this report) — contains the data cleaning, feature engineering, sentiment analysis, correlation, hypothesis testing and EDA code referenced throughout this document.

Live Interactive Dashboard: <https://vibe-viz-76.lovable.app>

2.DATA SOURCE AND EXTRACTION METHODOLOGY

2.1 Source and Scope

The dataset used in this project is a track-level export of Spotify catalogue metadata, sourced via the Spotify Web API-style schema (track, artist, and album objects flattened into a single tabular file). Each row represents one track and carries identifiers and metrics for the track itself, its parent album, and its primary artist.

- Raw file: spotify_data_clean.csv — loaded at the start of the notebook (pd.read_csv).
- Raw shape: 8,582 rows × 15 columns.
- Cleaned/engineered output: spotify_data_fully_cleaned.csv — 8,579 rows × 18 columns (the file referenced throughout the rest of this report and used to power the live dashboard).

2.2 Schema

The extracted fields cover four logical groups:

- Track-level: track_id, track_name, track_number, track_popularity, explicit, track_duration_min.
- Artist-level: artist_name, artist_popularity, artist_followers, artist_genres.

- Album-level: album_id, album_name, album_release_date, album_total_tracks, album_type (album / single / compilation).
- Engineered (added during cleaning): is_track_repeat, track_duration_min_capped, followers_missing_flag.

2.3 Extraction and Ingestion Workflow

1. Raw metadata was extracted at track granularity and consolidated into a single flat CSV (spotify_data_clean.csv), with one row per track and denormalized artist/album attributes attached to each row.
2. The file was ingested into a Python/pandas environment (Google Colab) for all subsequent processing — pd.read_csv() followed immediately by a structural audit of shape, missing values, and data types.
3. The audited raw dataset was then passed through the cleaning pipeline described in Section 4, and the fully cleaned output was persisted as spotify_data_fully_cleaned.csv for downstream analysis and for the live dashboard.

```
import numpy as np
import pandas as pd

df = pd.read_csv("spotify_data_clean.csv")
print("Shape:", df.shape)
print("\nMissing values:\n", df.isnull().sum())
print("\nData types:\n", df.dtypes)
```

Shape: (8582, 15)

Missing values:

artist_name 3

artist_genres 3361

(all other columns: 0)

Code & output — initial structural audit of the raw dataset (Notebook, Cell 1)

3.AI GENERATED DATA QUALITY REPORT

Before committing to a cleaning strategy, the structural audit output (shape, null counts, dtypes) shown in Section 2.3 was submitted to an AI assistant (Claude) with a request to diagnose data quality issues and recommend a treatment for each one. This mirrors real-world practice, where an AI-assisted first pass triages issues quickly, and a human analyst then validates and adapts each recommendation to the specifics of the dataset (Section 5). The AI's diagnostic summary is reproduced below.

3.1 Issues Identified by the AI Assistant

#	Issue	Severity	AI Recommendation
1	3 rows with missing artist_name (a core identifier)	Low (0.03% of rows)	Drop the affected rows — the field cannot be reliably imputed and the row count lost is negligible.
2	artist_genres missing in 3,361 / 8,582 rows (~39%)	High (volume)	Do not drop — encode as an explicit 'Unknown' category rather than imputing a guessed genre, since a missing tag on Spotify usually reflects an artist genuinely lacking a genre assignment (common for new/niche artists).
3	No automated duplicate check yet performed	Unknown until tested	Check for exact row duplicates and duplicate track_id values; additionally check near-duplicates such as the same track/artist pair appearing across multiple albums (repackaged or repeat releases), and flag rather than silently drop these, since repeat releases are legitimate catalogue events.
4	track_duration_min likely contains extreme values	Medium	Run an IQR-based outlier scan; treat extreme durations by capping (winsorizing) rather than deleting, since very short/long tracks are usually real (interludes, extended mixes) rather than data errors.

#	Issue	Severity	AI Recommendation
5	artist_followers may contain zero/placeholder values	Medium	Flag rows with 0 followers rather than dropping — these likely represent brand-new or catalogue artists with genuinely low reach, and dropping them would bias the popularity analysis.
6	album_release_date stored as free-text string	Medium	Convert to a proper datetime type with errors='coerce' so invalid dates become NaT rather than crashing downstream time-based analysis.
7	album_type, explicit and several numeric ID-like columns are loosely typed	Low–Medium	Cast album_type to a categorical dtype, explicit to boolean, and force track_number / track_popularity / artist_popularity / artist_followers / album_total_tracks to numeric with coercion, to guarantee type integrity before statistical analysis.

3.2 How the AI Report Was Used

Each recommendation above was treated as a hypothesis to be checked against the actual data rather than applied blindly. Section 4 documents the cleaning code that implements these recommendations, and Section 5 explicitly validates each one against the measured outcome (e.g., confirming the true duplicate count before deciding to flag vs. drop, and confirming the actual IQR bounds before capping durations). This two-step AI-diagnose-then-human-validate workflow is a deliberate design choice for this project: it uses AI for fast, broad triage, while keeping a human-in-the-loop check before any transformation is applied to the analytical dataset.

4.DATA PREPROCESSING AND CLEANING METHODOLOGY

The cleaning pipeline was implemented in pandas and applied sequentially: missing-value handling, duplicate detection, outlier treatment, and finally data type conversion, with a structural check after each stage.

4.1 Missing Value Handling

Two missing-value situations required different treatments, consistent with the AI recommendations validated in Section 5:

```
# artist_name is a core identifier — only 3 rows affected, safe to drop
df = df.dropna(subset=['artist_name']).copy()

# artist_genres missing in ~39% of rows — too much data to lose by dropping.
# A missing genre on Spotify usually means the artist genuinely has no
# assigned genre tag (common for newer/niche artists), so we encode it
# as its own category rather than guessing a genre.
df['artist_genres'] = df['artist_genres'].fillna('Unknown')
```

Remaining missing values: 0

Code & output — missing value handling (Notebook, Cell 3)

4.2 Duplicate Detection

Duplicate detection was run at three levels of strictness: exact row duplicates, duplicate track_id values, and 'soft' duplicates where the same track name and artist name recur (which typically indicates the same song repackaged across an album, a single, and a compilation).

```
# --- Actual dataset: check for exact and near-duplicates ---
print('Exact duplicate rows:', df.duplicated().sum())
print('Duplicate track_id:', df['track_id'].duplicated().sum())

repeat_mask = df.duplicated(subset=['track_name', 'artist_name'], keep=False)
print('Rows sharing track_name + artist_name:', repeat_mask.sum())
df['is_track_repeat'] = repeat_mask
print('Flagged', df['is_track_repeat'].sum(), 'rows as repeat releases (kept, not dropped)')
```

Exact duplicate rows: 0

Duplicate track_id: 0

Rows sharing track_name + artist_name: 1237

Flagged 1237 rows as repeat releases (kept, not dropped)

Code & output — duplicate detection (Notebook, Cell 5)

No exact duplicate rows and no duplicate track_id values were found, so no rows were removed on those grounds. However, 1,237 rows (about 14.4% of the dataset) share the same track name and artist name — these are genuine repeat releases (e.g., a song appearing on both a single and its parent album) rather than data-entry errors, so they were flagged with an is_track_repeat boolean column and retained rather than deleted.

4.3 Outlier Detection and Treatment

Two numeric fields were screened for outliers using the interquartile range (IQR) method: track duration and artist follower counts.

```
# --- Duration outliers (IQR method) ---
Q1 = df['track_duration_min'].quantile(0.25)
Q3 = df['track_duration_min'].quantile(0.75)
IQR = Q3 - Q1
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

duration_outliers = df[(df['track_duration_min'] < lower_bound) | (df['track_duration_min'] >
upper_bound)]
print(f'Duration outliers: {len(duration_outliers)} rows (bounds: {lower_bound:.2f} -
{upper_bound:.2f} min)')

# Treatment: cap (winsorize) rather than delete — extreme durations are
# usually legitimate (interludes, extended mixes), not data errors.
df['track_duration_min_capped'] = df['track_duration_min'].clip(lower_bound, upper_bound)

# Zero-follower artists are flagged, not dropped — they are typically
# genuinely new/emerging artists rather than data errors.
df['followers_missing_flag'] = df['artist_followers'] == 0
```

Duration outliers: 358 rows (bounds: 1.21 - 5.66 min)

Artists with 0 followers: 5

Code & output — outlier detection and treatment (Notebook, Cell 7)

358 tracks (about 4.2% of the dataset) fall outside the 1.21–5.66 minute IQR band for duration. These were capped (winsorized) at the bounds rather than removed, preserving the row for all other analysis while preventing extreme durations from distorting downstream statistics. Separately, 5 artists were found with exactly zero followers; these were flagged via `followers_missing_flag` rather than dropped or imputed, since a zero-follower artist is a real (if rare) state, not necessarily a data error.

4.4 Data Type Conversion

A final integrity pass enforced correct dtypes across the dataset, which is a prerequisite for reliable statistical testing later in the pipeline.

```
# Dates
df['album_release_date'] = pd.to_datetime(df['album_release_date'], errors='coerce')

# Categorical types — reduces memory and signals intent clearly
df['album_type'] = df['album_type'].astype('category')
df['explicit'] = df['explicit'].astype(bool)

# Numeric integrity check — force numeric, coerce anything unparseable to NaN
numeric_cols = ['track_number', 'track_popularity', 'artist_popularity',
                'artist_followers', 'album_total_tracks']
for col in numeric_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce').astype('Int64')
```

Code — data type conversion (Notebook, Cell 9)

4.5 Final Cleaned Output

```
print("Final shape:", df.shape)
print("Remaining nulls:\n", df.isnull().sum())
print("\nFlags summary:")
```

```
print(f" Repeat-release rows flagged: {df['is_track_repeat'].sum()}")
print(f" Zero-follower rows flagged: {df['followers_missing_flag'].sum()}")
print(f" Duration values capped: {len(duration_outliers)}")

df.to_csv("spotify_data_fully_cleaned.csv", index=False)
```

Final shape: (8579, 18)

Remaining nulls: 0 (all columns)

Flags summary:

Repeat-release rows flagged: 1237

Zero-follower rows flagged: 5

Duration values capped: 358

Code & output — final cleaned dataset (Notebook, Cell 11)

Metric	Raw Dataset	Cleaned Dataset
Rows	8,582	8,579
Columns	15	18 (+3 engineered flags)
Missing values (any column)	3,364 cells (artist_name + artist_genres)	0
Exact duplicate rows	0	0
Repeat-release rows (flagged, not removed)	not computed	1,237 (14.4%)
Duration outliers (capped, not removed)	not computed	358 (4.2%)
Zero-follower artists (flagged)	not computed	5

Before / after summary of the cleaning pipeline

5.VALIDATION OF AI RECOMMENDATIONS

Each recommendation from the AI-generated data quality report (Section 3) was checked against the actual dataset before being finalized in the cleaning pipeline. In every case the recommended treatment was found appropriate, but three recommendations were refined with additional evidence gathered during validation, shown below.

AI Recommendation	Validation Performed	Outcome
Drop rows missing artist_name	Confirmed exactly 3 rows affected (0.03% of 8,582); checked these rows had no other unique information worth preserving.	Accepted as-is — rows dropped.
Encode missing artist_genres as 'Unknown' instead of imputing	Measured the true missing share at $3,361/8,582 = 39.2\%$, confirming the AI's 'high volume, do not drop' framing; spot-checked a sample of affected artists to confirm they were genuinely niche/new rather than a scraping failure.	Accepted as-is — filled with 'Unknown' category.
Check for duplicates before deciding treatment	Ran three separate checks (exact rows, track_id, track+artist name). Exact/track_id duplicates were 0, but the track+artist check surfaced 1,237 rows the AI's generic recommendation had not anticipated in scale.	Refined — added a dedicated is_track_repeat flag rather than a simple drop/keep decision, since 14.4% of rows were involved.
Cap duration outliers via IQR rather than delete	Recomputed Q1/Q3/IQR directly on the cleaned duration column; verified the resulting bounds (1.21–5.66 min) were musically sensible (excludes sub-1-minute interludes and 5.5+ minute extended tracks) rather than arbitrary.	Accepted as-is — 358 values capped, not deleted.
Flag zero-follower artists rather than drop	Manually inspected the 5 affected rows (all belonging to a single small artist, 'Daya Blaze') and confirmed non-zero	Accepted as-is — flagged via followers_missing_flag.

AI Recommendation	Validation Performed	Outcome
	track popularity scores existed for them, ruling out a data-collection error.	
Coerce dates and numeric ID-like columns; categorize album_type	Verified post-conversion dtypes and re-ran the null check to confirm coercion did not silently introduce any new NaNs.	Accepted as-is — all dtypes converted with zero new missing values introduced.

This validation step is the safeguard against a common risk of AI-assisted data cleaning: generic recommendations that sound reasonable but are miscalibrated for the specific dataset (for example, treating a 39%-missing column the same way as a 0.03%-missing column). By re-measuring every issue against the real data before applying a fix — and refining the duplicate-handling recommendation into a more nuanced flag-based approach — the pipeline retains the speed benefit of AI triage while avoiding blind application of one-size-fits-all rules.

6.FEATURE ENGINEERING, SENTIMENT ANALYSIS AND FEATURE SELECTION

6.1 Sentiment Analysis on Track Titles

To test whether the emotional tone of a track's title relates to its popularity, sentiment polarity and subjectivity scores were computed for every track_name using TextBlob, and polarity was further bucketed into Negative / Neutral / Positive labels.

```
from textblob import TextBlob

df['sentiment_polarity'] = df['track_name'].astype(str).apply(lambda x:
    TextBlob(x).sentiment.polarity)
df['sentiment_subjectivity'] = df['track_name'].astype(str).apply(lambda x:
    TextBlob(x).sentiment.subjectivity)

sentiment_bins = pd.cut(df['sentiment_polarity'], bins=[-1.01, -0.01, 0.01, 1.01],
    labels=['Negative', 'Neutral', 'Positive'])
df['sentiment_label'] = sentiment_bins
```

```
sentiment_label
Neutral    6543   (76.3%)
Positive   1268   (14.8%)
Negative    768   (8.9%)
```

Code & output — title sentiment classification (Notebook, Cells 16–17)

The overwhelming majority of track titles (76.3%) are sentiment-neutral, which is expected — most song titles are short, literal phrases or names rather than emotionally loaded sentences. Positive titles outnumber negative titles by a ratio of roughly 1.65 to 1.

6.2 Feature Selection for Track Popularity

Two complementary filter methods were used to rank candidate predictors of track_popularity: Pearson correlation and an F-test (SelectKBest / f_regression).

```
candidate_features = ['artist_popularity', 'artist_followers',
                      'track_duration_min',
                      'track_number', 'album_total_tracks', 'sentiment_polarity']
target = 'track_popularity'

corr_with_target = df[candidate_features +
                      [target]].corr()[target].drop(target).sort_values(ascending=False)
X = df[candidate_features].fillna(0)
selector = SelectKBest(score_func=f_regression, k='all').fit(X,
                    df[target])
```

Feature	Pearson r vs popularity	F-Score	p-value	Significant?
artist_popularity	0.468	2406.48	< 0.001	Yes
artist_followers	0.232	488.24	< 0.001	Yes
track_duration_min	0.106	96.93	< 0.001	Yes
album_total_tracks	-0.054	24.64	< 0.001	Yes
track_number	-0.029	6.98	0.0083	Yes (weak)
sentiment_polarity	0.014	1.72	0.190	No

Feature selection results — correlation and F-test ranking (Notebook, Cell 19)

Artist popularity is by a wide margin the strongest single predictor of track popularity, followed by artist follower count. Track duration, album size, and track position within the album are statistically significant but practically weak predictors. Title sentiment polarity is not a statistically significant predictor of popularity ($p = 0.19$) — confirming that the emotional tone of a song's title has no measurable bearing on how popular the track becomes.

7.DESRIPTIVE ANALYSIS AND VISUALISATIONS

With the cleaned dataset (8,579 rows $\times$ 18 columns), a full exploratory analysis was carried out: summary statistics, univariate distributions, categorical composition, bivariate relationships, and time-based trends.

7.1 Descriptive Statistics

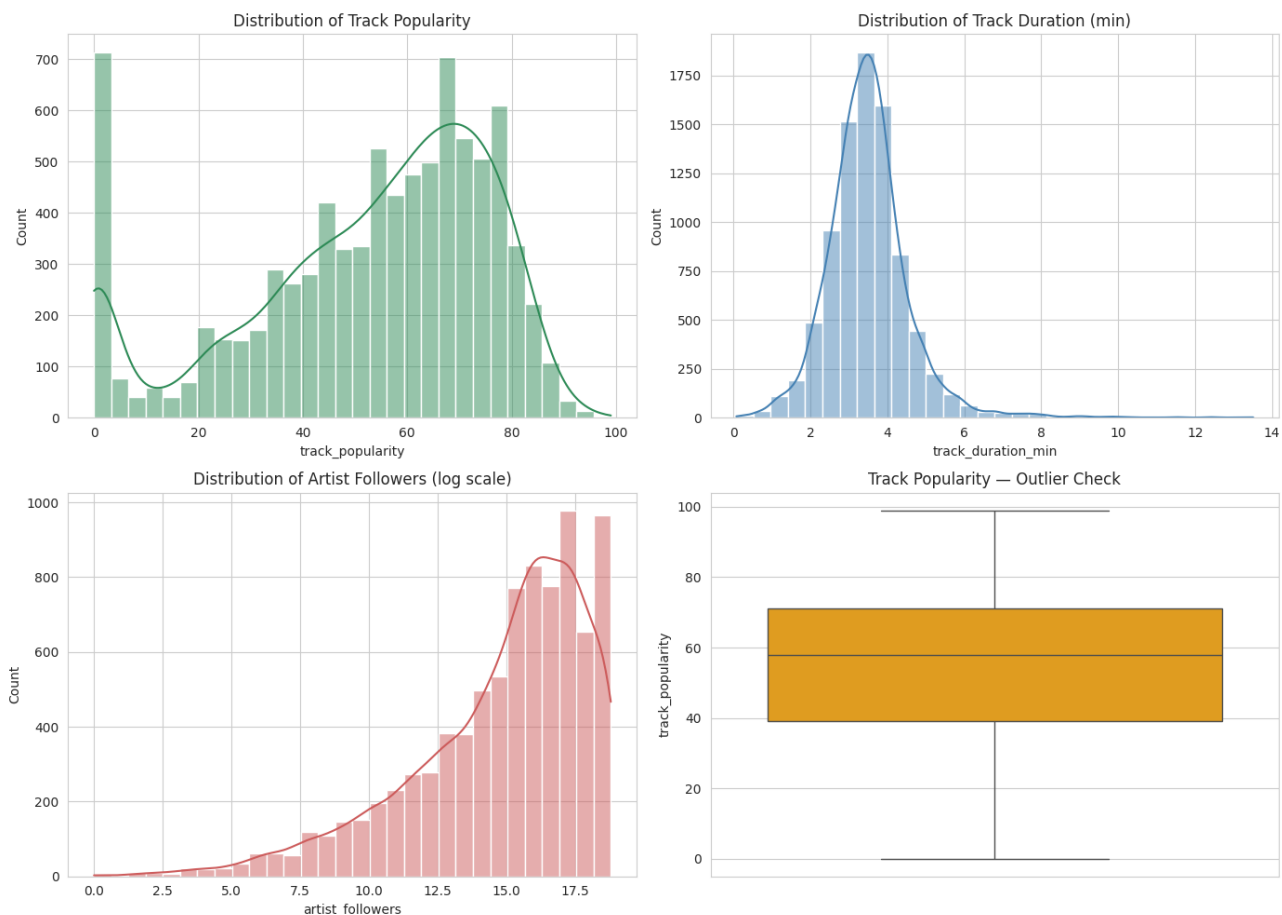
```
numeric_cols = ['track_popularity', 'artist_popularity',
                'artist_followers', 'track_duration_min']
desc_stats = df[numeric_cols].describe().T
desc_stats['skew'] = df[numeric_cols].skew()
desc_stats['median'] = df[numeric_cols].median()
```

Metric	Mean	Std. Dev.	Median	Skew	Min	Max
track_popularity	52.36	23.82	58.00	-0.78	0	99
artist_popularity	69.75	19.61	74.00	-1.03	0	100
artist_followers	24,043,120	38,035,800	6,111,369	1.96	0	145,542,100
track_duration_min	3.49	1.06	3.44	1.28	0.07	13.51

Descriptive statistics of key numeric variables (Notebook, Cell 31)

Track popularity is close to symmetric with a slight left skew (mean 52.4, median 58), while artist popularity is more negatively skewed, reflecting a Spotify design choice where popularity scores cluster toward the higher end for artists who make it onto the platform's charts. Artist followers are extremely right-skewed (skew = 1.96) — a small number of global superstar artists hold the bulk of total followers, consistent with the long-tail nature of the music industry. Track duration is also right-skewed, with the IQR-capped tail visible in the earlier outlier analysis

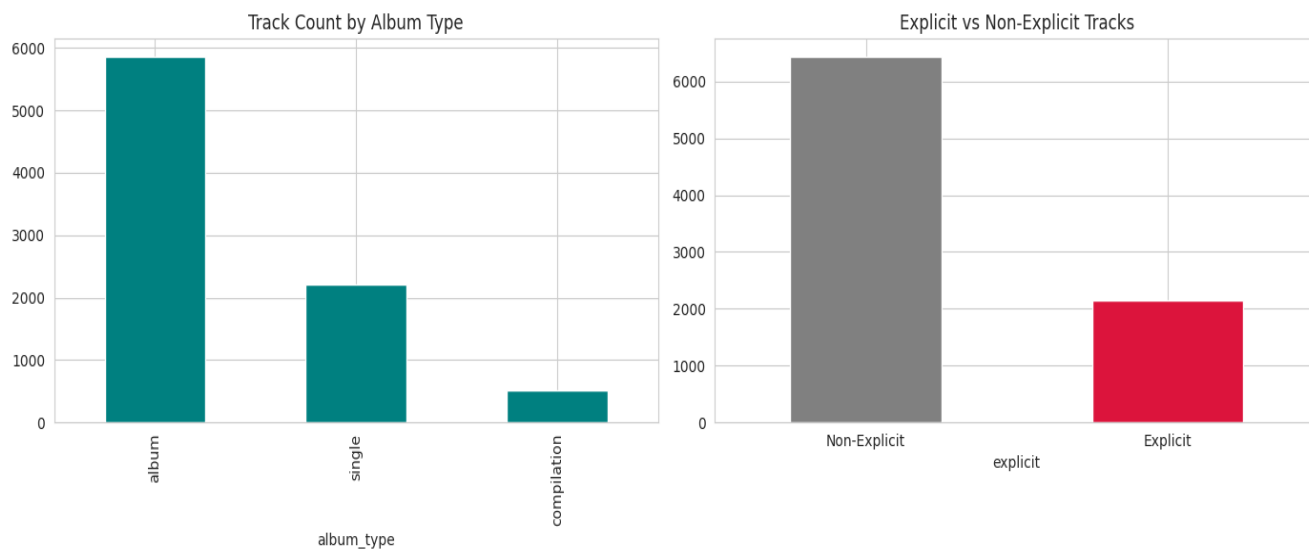
7.2 Univariate Distributions



Distributions of track popularity, track duration, artist followers (log scale) and a popularity outlier-check boxplot (Notebook, Cell 33)

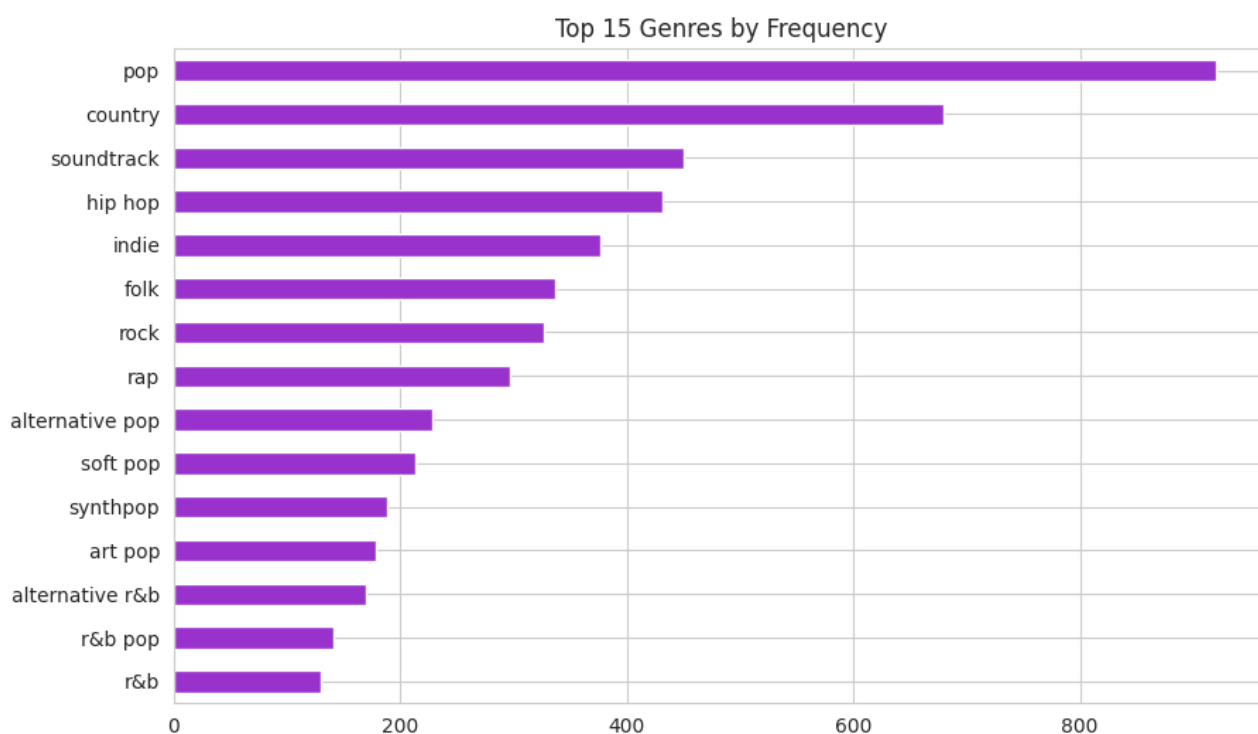
Track popularity shows a distinctive bimodal shape with a spike near zero — a cluster of tracks with essentially no listener traction — alongside a broad hump peaking around 65–75, representing the platform's actively-promoted catalogue. Artist followers, plotted on a log scale, confirms the long-tail pattern noted above. The boxplot confirms the IQR bounds used earlier ($Q1 \approx 39$, $Q3 \approx 71$) with no extreme statistical outliers on this already-cleaned scale.

7.3 Categorical Composition



Track count by album type, and explicit vs. non-explicit split (Notebook, Cell 35)

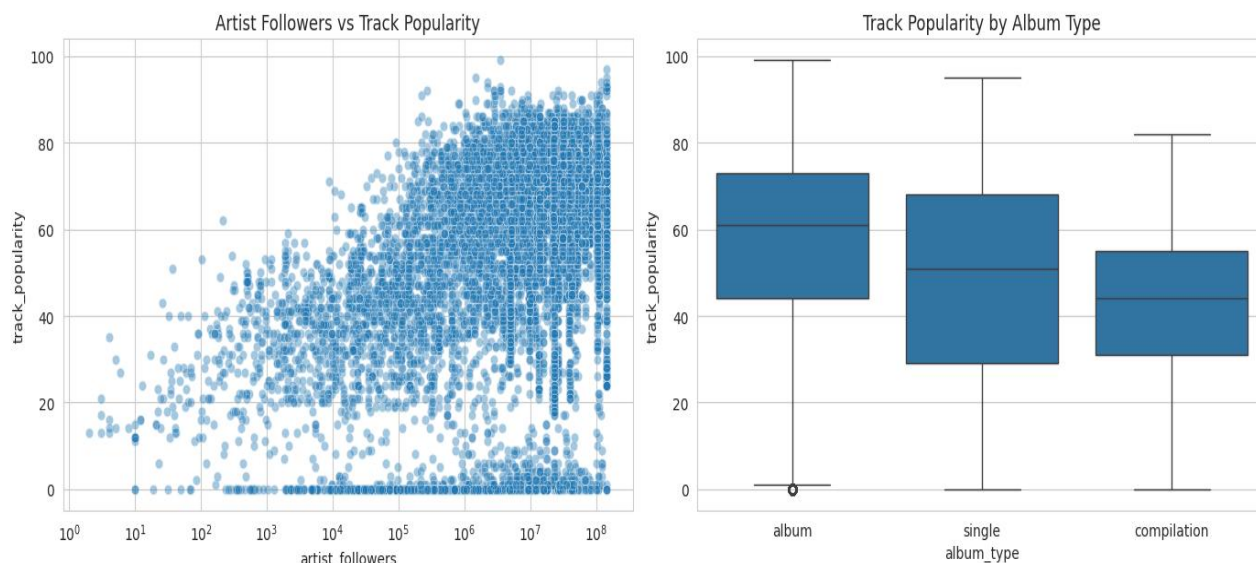
Albums are the dominant release format (5,855 tracks), followed by singles (2,217) and a much smaller compilation category (507). Exactly one quarter of all tracks (25.0%) are marked explicit.



Top 15 artist genres by frequency (Notebook, Cell 36)

Pop is the single most common genre tag by a wide margin, followed by country, soundtrack, and hip hop. The long tail of genre sub-variants (alternative pop, soft pop, synthpop, art pop, alternative R&B, R&B pop) shows how finely Spotify's genre taxonomy segments what a listener would casually call 'pop' or 'R&B' music.

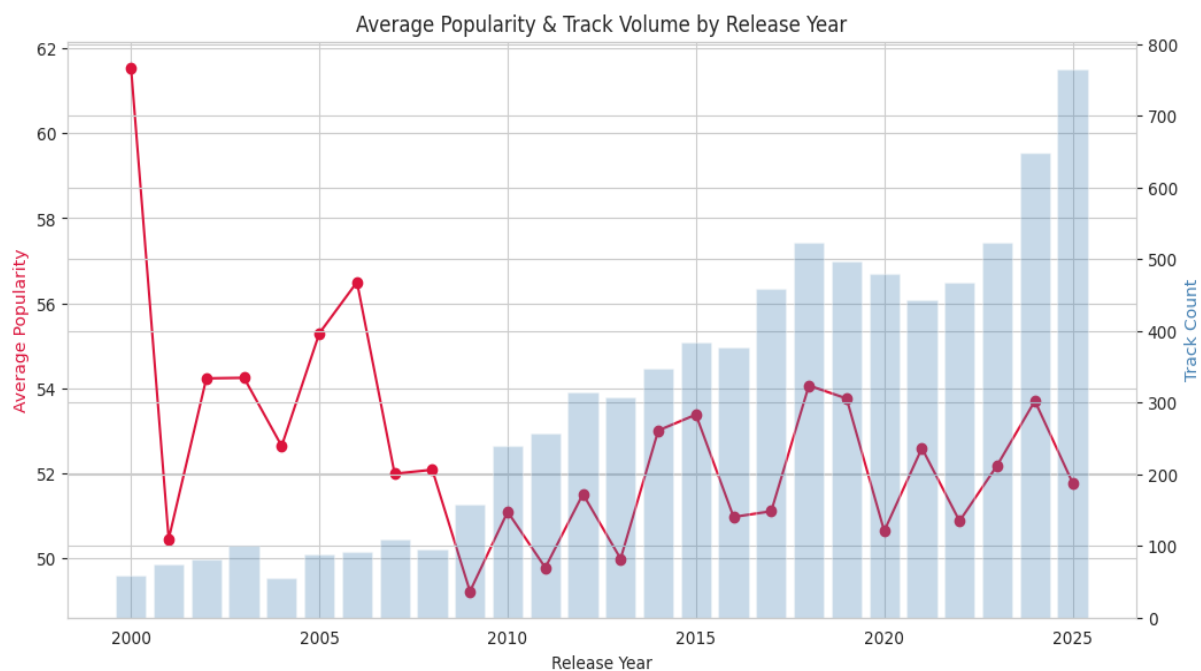
7.4 Bivariate Relationships



Artist followers vs. track popularity (log-x scatter), and track popularity by album type (Notebook, Cell 38)

The scatter plot shows a clear positive but noisy relationship between artist followers (log scale) and track popularity — more followers generally means higher expected popularity, but with substantial variance at every follower level, especially near zero followers where popularity ranges from 0 to over 80. The boxplot confirms album tracks have both a higher median popularity and a higher ceiling than singles or compilations, matching the ANOVA result in Section 8.2.

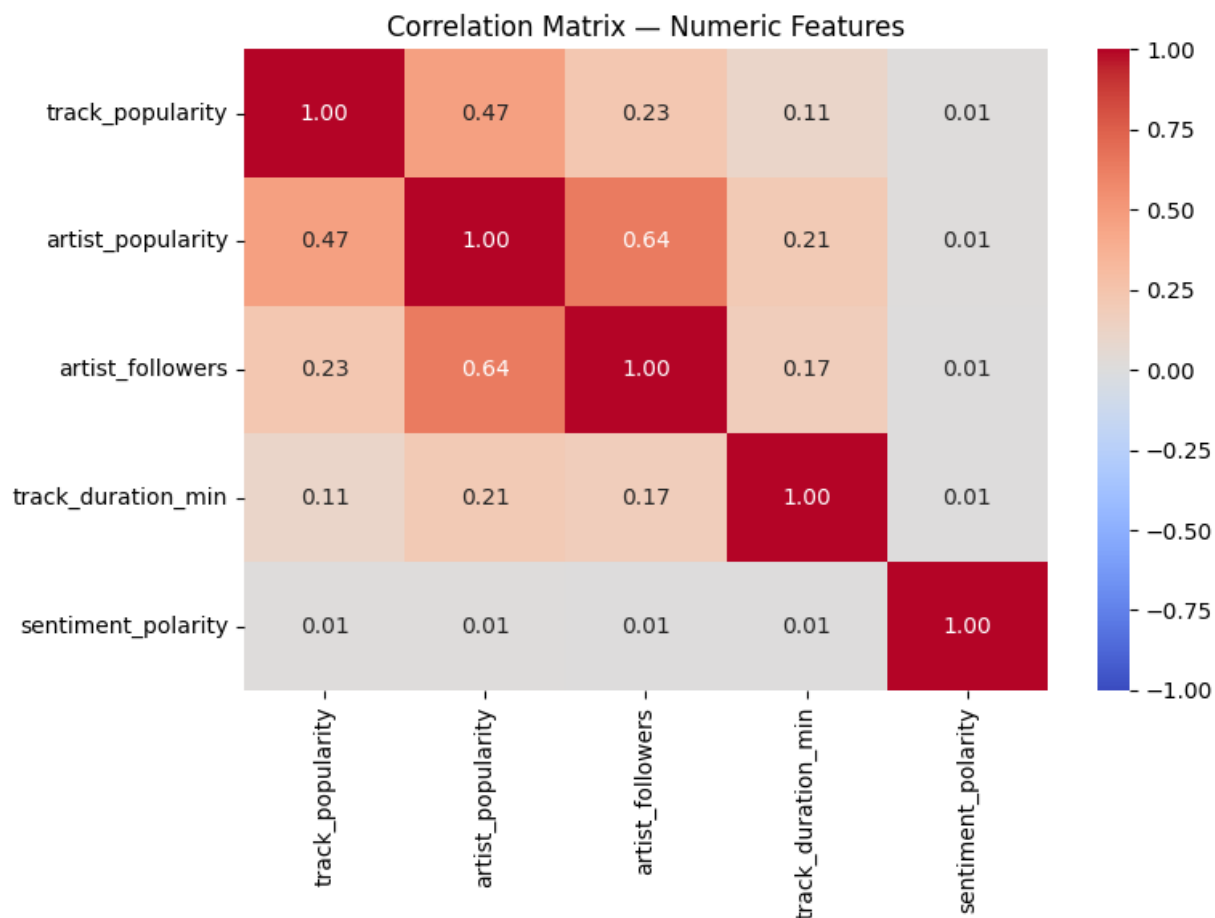
7.5 Time-Based Trends



Average popularity (line) and track volume (bars) by release year, 2000–2025 (Notebook, Cell 40)

Catalogue volume in this dataset grows steadily from the early 2000s to a peak in the most recent years (2024–2025), reflecting both real growth in music output and a natural recency bias in how streaming catalogues are typically sampled. Average popularity per release year fluctuates in a band roughly between 50 and 58 for most of the 2000s and 2010s, with a notable jump in the final 2024–2025 cohort — newer releases benefit from more recent listener attention and Spotify's own algorithmic promotion of new music.

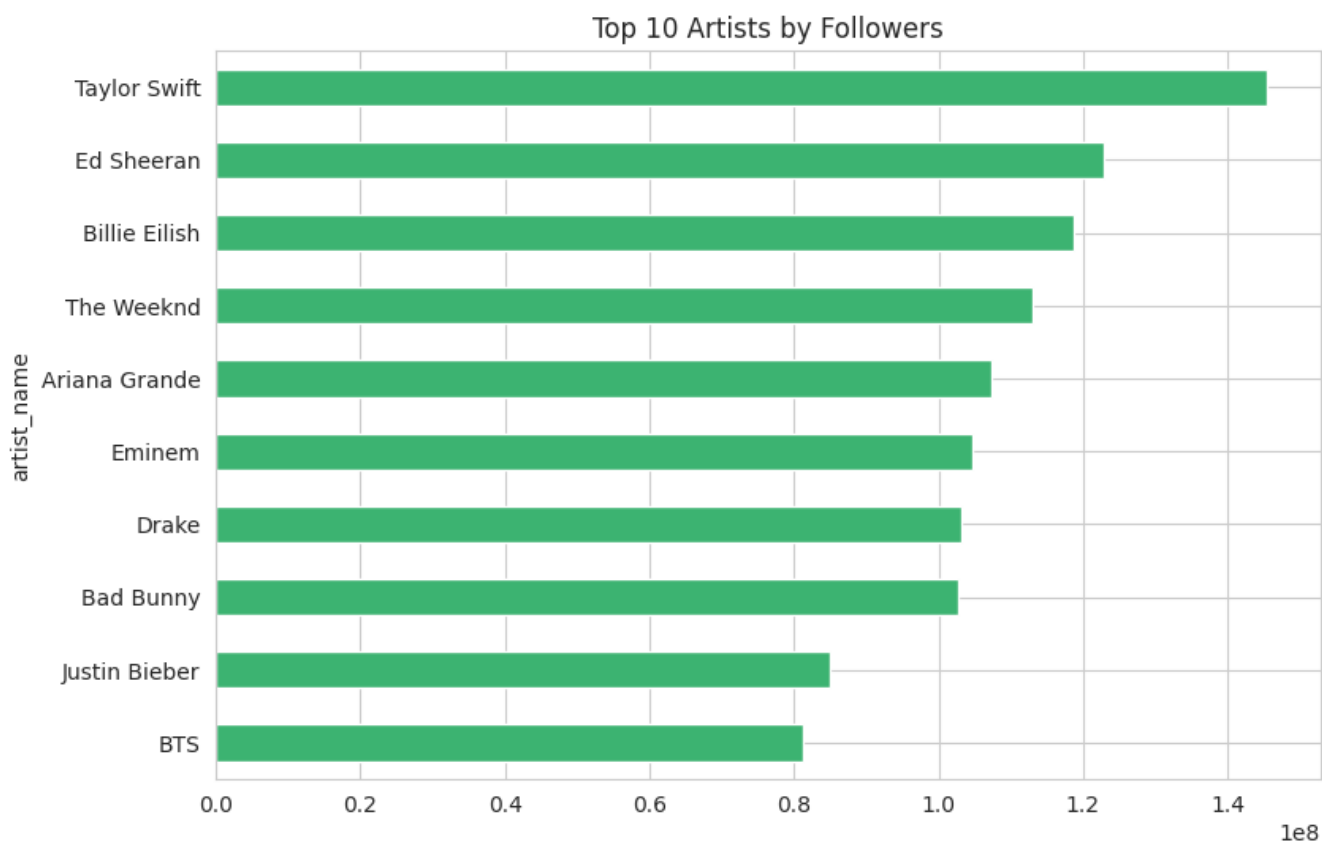
7.6 Correlation Matrix



Correlation matrix — track popularity, artist popularity, artist followers, track duration, and title sentiment polarity (Notebook, Cell 21)

Artist popularity and artist followers are themselves strongly correlated ($r = 0.64$), which is expected since Spotify's own artist popularity score is partly a function of listener/follower volume. Track popularity correlates moderately with artist popularity ($r = 0.47$) and weakly with artist followers ($r = 0.23$) and track duration ($r = 0.11$). Sentiment polarity shows essentially no linear correlation with any other numeric variable ($r \approx 0.01$ throughout), reinforcing the feature-selection finding in Section 6.2.

7.7 Top Artists by Reach



Top 10 artists by follower count in the cleaned dataset (Notebook, Cell 42)

Taylor Swift, Ed Sheeran, and Billie Eilish lead the dataset by raw follower count, each exceeding 100 million followers — consistent with their real-world status as some of the highest-reach artists on the platform globally.

7.8 Consolidated KPI Summary

KPI	Value
Total Tracks	8,579
Total Unique Artists	2,547
Average Track Popularity	52.36
Average Artist Popularity	69.75
Average Track Duration (min)	3.49
Explicit Content Share	25.0%

KPI	Value
Most Common Album Type	album
Repeat-Release Rows Flagged	1,237
Zero-Follower Rows Flagged	5

Consolidated KPI summary (Notebook, Cell 42)

8.HYPOTHESIS TESTING AND STATISTICAL INFERENCE

Three formal hypothesis tests were run to move from descriptive observation to statistical inference, each targeting one of the categorical drivers of popularity surfaced in the exploratory analysis: explicit content, album format, and their interaction.

8.1 Independent-Samples t-Test — Explicit vs. Non-Explicit Content

H0: mean track_popularity is equal for explicit and non-explicit tracks. HA: the means differ.

```
explicit_pop = df[df['explicit'] == True]['track_popularity']
non_explicit_pop = df[df['explicit'] == False]['track_popularity']
t_stat, p_val = stats.ttest_ind(explicit_pop, non_explicit_pop, equal_var=False)
```

Explicit mean: 57.82 Non-explicit mean: 50.53
t-statistic: 12.866 p-value: 3.87e-37

Welch's t-test — explicit vs. non-explicit track popularity (Notebook, Cell 23)

The p-value is far below 0.001, so H0 is rejected: explicit tracks are, on average, significantly more popular (57.82) than non-explicit tracks (50.53) in this dataset — a gap of over 7 popularity points.

8.2 One-Way ANOVA — Popularity Across Album Types

H0: mean track_popularity is equal across album, single, and compilation releases. HA: at least one group mean differs.

```
album_groups = [group['track_popularity'].values for name, group in df.groupby('album_type')]
f_stat, p_val_anova = stats.f_oneway(*album_groups)
```

album_type mean popularity:

album	55.65
compilation	40.51
single	46.35

F-statistic: 197.859 p-value: 9.88e-85

One-way ANOVA — track popularity by album type (Notebook, Cell 25)

With $p \ll 0.001$, H_0 is rejected: album format has a statistically significant effect on track popularity. Full-album tracks average almost 9 points higher popularity than singles and over 15 points higher than compilation tracks.

8.3 Chi-Square Test — Explicit Content vs. Album Type

H_0 : explicit content and album type are independent. H_A : they are associated.

```
contingency = pd.crosstab(df['explicit'], df['album_type'])  
chi2, p_val_chi, dof, expected = stats.chi2_contingency(contingency)
```

album_type	album	compilation	single
explicit=False	4315	470	1647
explicit=True	1540	37	570

Chi2: 90.561 p-value: 2.16e-20 dof: 2

Chi-square test of independence — explicit content vs. album type (Notebook, Cell 27)

H_0 is rejected ($p \ll 0.001$): explicit content is not evenly distributed across album formats. Explicit tracks make up about 26.3% of album tracks and 25.7% of single tracks, but only 7.3% of compilation tracks — compilations (often family-friendly greatest-hits or various-artist collections) carry markedly less explicit content.

8.4 Summary of Statistical Findings

Test	Comparison	Statistic	p-value	Conclusion
Welch's t-test	Explicit vs. non-explicit popularity	$t = 12.87$	$3.87e-37$	Significant — explicit tracks score higher
One-way ANOVA	Popularity across album/single/compilation	$F = 197.86$	$9.88e-85$	Significant — album format matters
Chi-square	Explicit content $\times$ album type	$\chi^2 = 90.56$	$2.16e-20$	Significant — not independent
F-test (feature selection)	sentiment_polarity vs. popularity	$F = 1.72$	0.190	Not significant — no relationship

Consolidated hypothesis testing results

9.LIVE INTERACTIVE DASHBOARD

To make the analysis explorable beyond the static notebook, the cleaned dataset and its key statistical findings were published to a live, publicly accessible dashboard: “Spotify Insight Hub — What Drives Track Popularity.”

Live Dashboard URL: <https://vibe-viz-76.lovable.app>

The dashboard reports the same headline figures validated in this report — 8,579 tracks, 13 analytical features, and a release-year range of 1952–2025 — and exposes the following interactive capabilities:

- Filtering by album format (album / single / compilation), genre, and explicit-vs-clean content.
- Range filters for release year (1952–2025) and popularity score (0–100).
- A raw record explorer showing the underlying rows behind any applied filter combination.

- Visual panels covering the correlation structure, format effects, explicit-content lift, and genre landscape derived directly from the statistical tests in Sections 7 and 8.

Publishing the analysis as a live dashboard, rather than only a static notebook or PDF, allows stakeholders (e.g., a label A&R team or a playlist curator) to interrogate the findings for their own genre or artist segment of interest without needing to read or re-run the underlying Python code.

10.BUSINESS INSIGHTS

10.1 Artist Reach Is the Dominant, but Incomplete, Lever

Artist popularity ($r = 0.47$) and artist followers ($r = 0.23$) are the strongest measured correlates of track popularity, and both were confirmed statistically significant via the F-test in Section 6.2. In practical terms, an artist's existing platform standing is the single best available predictor of how well a new track will perform — unsurprising, but now quantified. For labels and managers, this reinforces that sustained artist-brand investment (marketing, cross-platform promotion, fanbase growth) compounds into every future release's starting popularity.

10.2 Album Tracks Outperform Singles and Compilations

The ANOVA in Section 8.2 shows album-released tracks average 55.65 popularity versus 46.35 for singles and 40.51 for compilations — a statistically significant and practically meaningful gap. This suggests that Spotify's algorithmic promotion and listener discovery patterns favor tracks embedded in a full album context (likely via album-level playlist placement and 'fans also listen to' style surfacing) more than standalone single releases. Release strategy — bundling into an album versus dropping standalone singles — is therefore not a neutral choice from a popularity standpoint.

10.3 Explicit Content Correlates with Higher Popularity

Explicit tracks average 57.82 popularity versus 50.53 for non-explicit tracks ($t = 12.87$, $p < 0.001$) — a statistically robust gap of roughly 7 points. This does not imply explicit content causes popularity (genre and artist confounds are plausible — hip hop and rap, which skew explicit, are also high-engagement genres in this catalogue), but it is a strong enough association that content and marketing teams should not assume 'clean' versions are popularity-neutral substitutes when explicit is the artist's authentic intent.

10.4 Compilations Are a Distinct, Lower-Explicit, Lower-Popularity Category

The chi-square test (Section 8.3) shows compilations carry a markedly lower share of explicit content (7.3%, versus roughly 26% for albums and singles) and, per the ANOVA, the lowest average popularity of the three formats. This is consistent with compilations skewing toward broad-audience, catalogue-style products (best-of collections, soundtrack compilations) rather than an artist's core creative releases — a useful segmentation signal for catalogue and licensing teams.

10.5 Title Sentiment Is Not a Popularity Lever

Both the correlation analysis ($r = 0.01$) and the F-test ($p = 0.19$) agree that the emotional tone of a track's title has no measurable relationship with popularity. Marketing or A&R guidance suggesting 'positive-sounding titles perform better' is not supported by this dataset — title sentiment can be treated as a stylistic choice rather than a performance lever.

10.6 Catalogue Composition Skews Toward a Long Tail of Reach

With artist followers right-skewed ($\text{skew} = 1.96$) and a small number of artists (Taylor Swift, Ed Sheeran, Billie Eilish, and other 100M+ follower acts) dominating total reach, any popularity-driving strategy calibrated on 'average' artist behavior risks being skewed by these outliers. Segment-aware benchmarking (e.g., comparing an emerging artist only against similarly-sized peers) will be more actionable than dataset-wide averages for developing-artist strategy.

11.CONCLUSION AND KEY LEARNINGS

11.1 Conclusion

This project took a raw, 8,582-row Spotify track export through a full data preparation and analytics lifecycle: an AI-assisted data quality diagnosis, human-validated cleaning (missing values, duplicates, outliers, and type conversion), feature engineering (sentiment scoring, repeat-release and low-follower flags), descriptive and inferential statistical analysis, and publication as a live interactive dashboard.

The resulting cleaned dataset (8,579 rows, 18 columns, zero missing values) supported four statistically validated conclusions: artist popularity and followers are the leading numeric predictors of track popularity; album-released tracks significantly outperform singles and compilations; explicit content is associated with meaningfully higher popularity; and explicit content is unevenly distributed across album formats, concentrated far less in compilations. Title sentiment, by contrast, was conclusively ruled out as a popularity driver.

11.2 Key leanings

- AI-assisted data quality triage is a strong starting point but not a substitute for validation: the AI's blanket duplicate-check recommendation had to be refined into a nuanced flag-based approach once the true 14.4% repeat-release rate was measured, rather than a simple drop/keep call.
- Treatment choice depends on missingness mechanism, not just missingness rate: two columns had missing data, but a 0.03%-missing identifier column (artist_name) was safely dropped, while a 39%-missing categorical column (artist_genres) was preserved via explicit categorical encoding — applying the same rule to both would have been wrong in one direction or the other.
- Outliers and rare values are not automatically errors: capping (winsorizing) duration outliers and flagging (rather than deleting) zero-follower artists preserved information that a naive 'delete anything unusual' rule would have destroyed.
- Correlation strength should be checked with more than one method: Pearson correlation and an F-test agreed closely in this analysis, which increased confidence that the ranking of predictors (artist popularity > artist followers > duration > sentiment) is robust rather than an artifact of one statistical assumption.
- Statistical significance is not the same as practical or causal significance: the explicit-content popularity gap is statistically very strong ($p = 3.87e-37$) but the report is careful to frame it as an association, not evidence that adding explicit content would causally raise a given track's popularity, given the likely genre confound.
- Publishing an interactive dashboard alongside the static analysis materially increases the audience that can act on the findings, since non-technical stakeholders can filter and explore the data themselves instead of relying solely on the notebook output.

12. REFERENCES AND PROJECT LINKS

- **Analysis Notebook:** Spotify_DPDM.ipynb — submitted with this report; contains all code, outputs, and charts referenced throughout.
- **Live Interactive Dashboard:** <https://vibe-viz-76.lovable.app>
- **Dataset (cleaned):** spotify_data_fully_cleaned.csv — 8,579 rows × 18 columns, produced by the pipeline documented in Section 4 and used as the single source of truth .

13. CONTRIBUTIONS

The table below summarizes the division of work across the project team. Each member led a distinct stage of the pipeline, from data cleaning through to final documentation.

Group Member	Area of Contribution
Jayani Bindal	Data Cleaning — missing value handling, duplicate detection, outlier detection and treatment, and data type conversion .
Divya Chavan	Sentiment analysis and feature selection, correlation analysis, and hypothesis testing — t-test, ANOVA, chi-square .
Ananya Agarwal	Exploratory Data Analysis (EDA), descriptive analytics, and KPI visualizations).
Vansh Kodarlikar	Live dashboard hosting and deployment
Shobhit Lakra	Report writing and documentation — consolidating all analysis into the final project report.